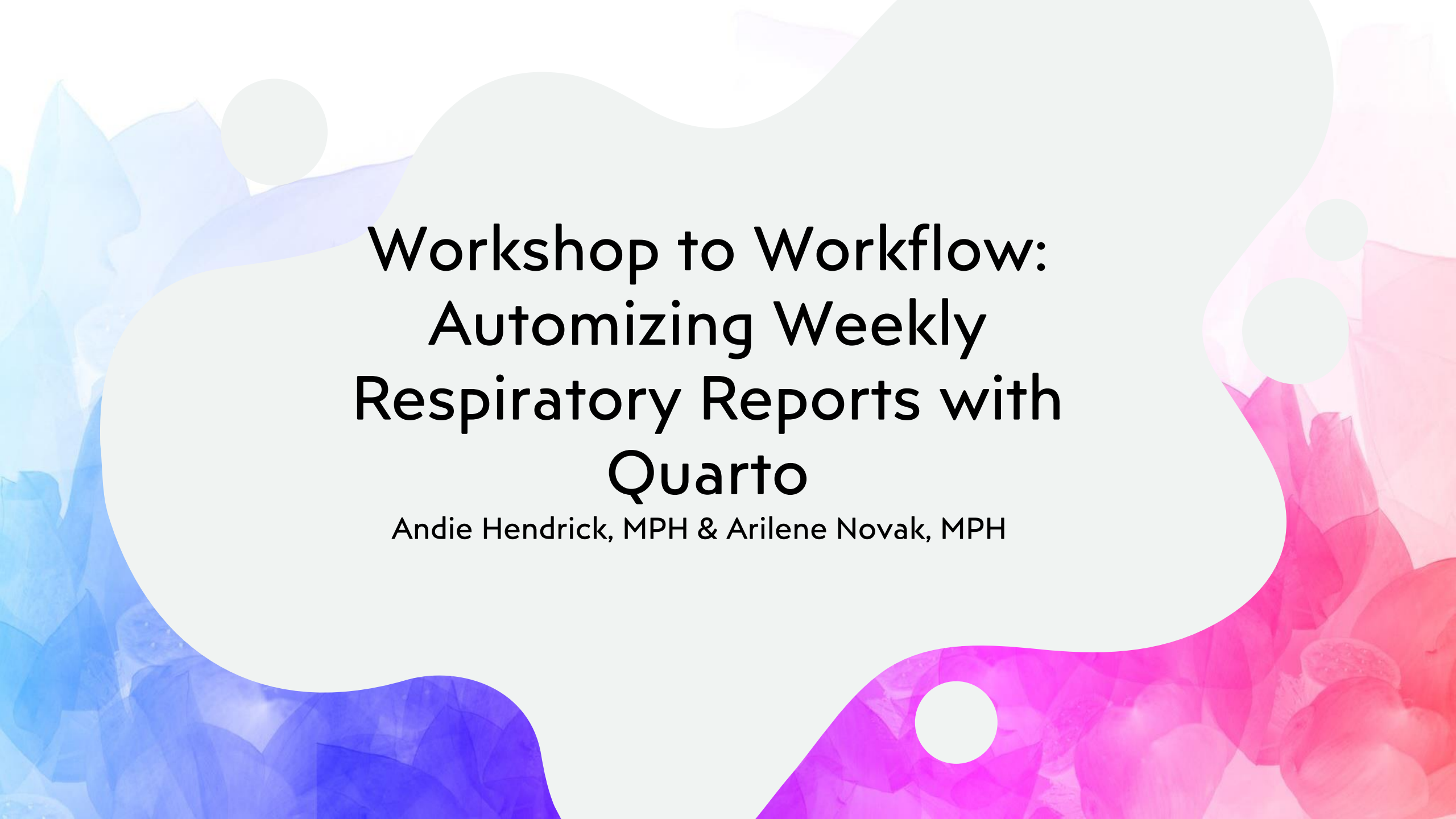


The background features a central white organic shape containing the text. This shape is surrounded by vibrant, abstract patterns: blue and purple leaf-like shapes on the left, and pink and red floral-like shapes on the right. Several light green circles of varying sizes are scattered throughout the composition.

# Workshop to Workflow: Automizing Weekly Respiratory Reports with Quarto

Andie Hendrick, MPH & Arilene Novak, MPH

# A year in the life of a respiratory epidemiologist

## October - April

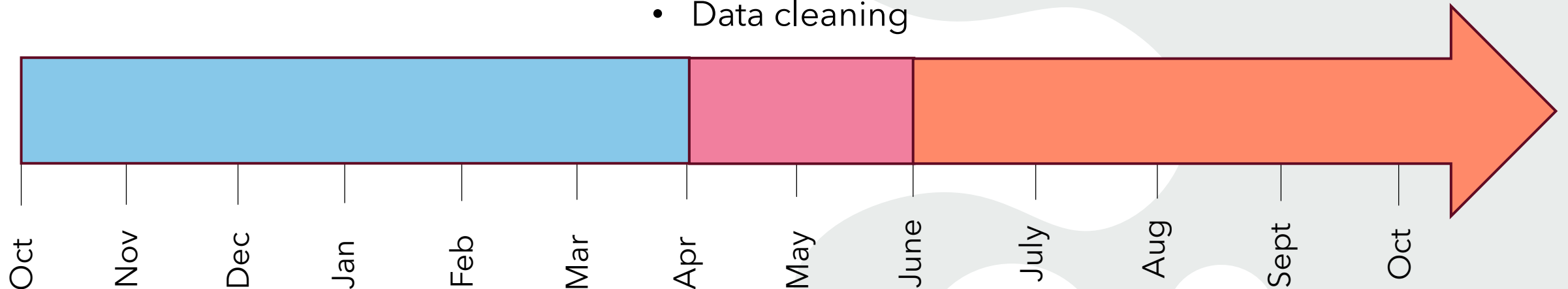
- Publication of weekly public facing reports
- Hospitalization case finding
- Hospitalization chart reviews
- Weekly data transmission to CDC

## July - September

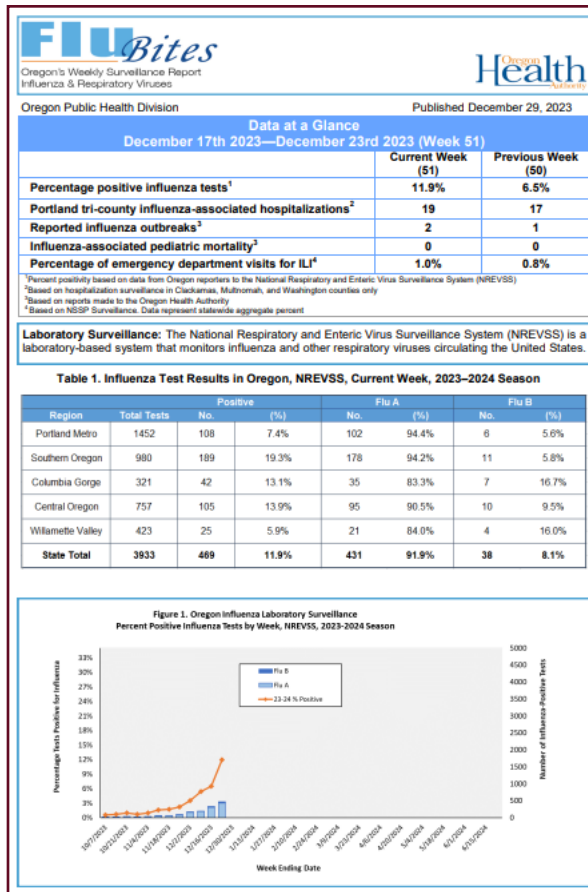
- Data modernization projects

## May - June

- CDC special projects
- Data cleaning



# Weekly reporting: old process

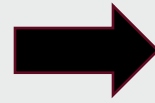


**Data at a Glance**  
**December 17th 2023—December 23rd 2023 (Week 51)**

|                                                                        | Current Week (51) | Previous Week (50) |
|------------------------------------------------------------------------|-------------------|--------------------|
| Percentage positive influenza tests <sup>1</sup>                       | 11.9%             | 6.5%               |
| Portland tri-county influenza-associated hospitalizations <sup>2</sup> | 19                | 17                 |
| Reported influenza outbreaks <sup>3</sup>                              | 2                 | 1                  |
| Influenza-associated pediatric mortality <sup>3</sup>                  | 0                 | 0                  |
| Percentage of emergency department visits for ILI <sup>4</sup>         | 1.0%              | 0.8%               |

<sup>1</sup>Percent positivity based on data from Oregon reporters to the National Respiratory and Enteric Virus Surveillance System (NREVSS)  
<sup>2</sup>Based on hospitalization surveillance in Clackamas, Multnomah, and Washington counties only  
<sup>3</sup>Based on reports made to the Oregon Health Authority  
<sup>4</sup>Based on NSSP Surveillance. Data represent statewide aggregate percent

# Weekly reporting: new process



|                                           |                     |        |
|-------------------------------------------|---------------------|--------|
| R 1 EIP Cleaning                          | 5/28/2025 4:48 PM   | R File |
| R 2 EIP Summary Table_FlexTable           | 9/26/2024 3:55 PM   | R File |
| R 3 EIP Hospitalization Graph_Rates       | 5/28/2025 4:48 PM   | R File |
| R 4 Epi Weeks                             | 9/23/2024 10:33 AM  | R File |
| R 5 NREVSS Cleaning                       | 10/23/2024 1:41 PM  | R File |
| R 6 NREVSS Multiseason Graph              | 4/7/2025 2:07 PM    | R File |
| R 6.5 Providence Code_V2                  | 1/22/2025 12:13 PM  | R File |
| R 7 NREVSS Regional Summary Table         | 1/8/2025 11:53 AM   | R File |
| R 8 Dual Y Axis PP and Testing Counts     | 1/2/2025 9:43 AM    | R File |
| R 9 Outbreaks Cleaning                    | 9/20/2024 12:47 PM  | R File |
| R 10 Outbreaks Graph                      | 9/20/2024 12:47 PM  | R File |
| R 11 Flu Pediatric Death Count            | 9/9/2024 2:15 PM    | R File |
| R 12 Flu Bites Summary Table_FlexTable_V2 | 2/12/2025 11:25 AM  | R File |
| R 13 FluBites Text Values                 | 10/16/2024 10:06 AM | R File |
| R 14 PP Map                               | 1/29/2025 10:33 AM  | R File |

- 15 individual scripts
- 1 QMD file
- 1 template with agency logo

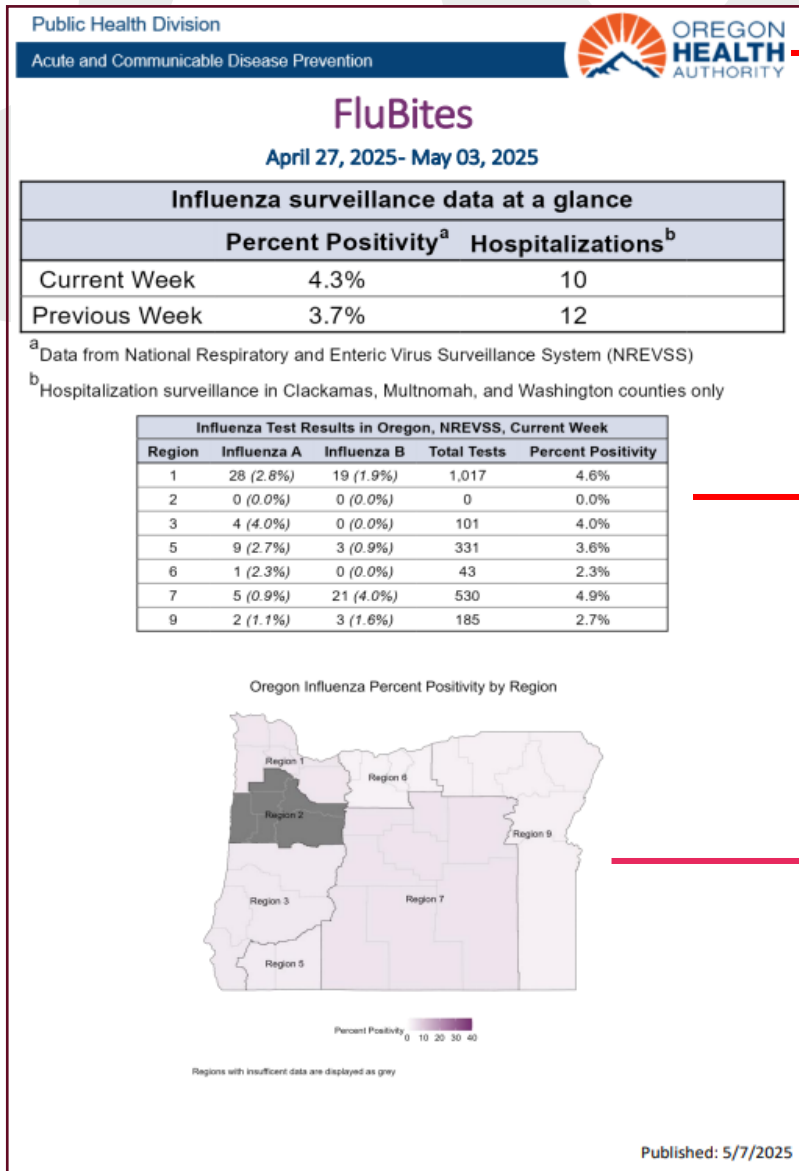
```
{r}
#| label: run scripts 1
#| eval: TRUE
#| echo: FALSE
#| message: FALSE
#| warning: FALSE
#| include: FALSE

#Data Import and Cleaning
source("I:/ACDP-Respiratory/FluBites/R code/2024-2025/1 EIP Cleaning.R")

#Data Summary Table
source("I:/ACDP-Respiratory/FluBites/R code/2024-2025/2 EIP Summary Table_FlexTable.R")

#Hospitalization rates graph: ggplot
source("I:/ACDP-Respiratory/FluBites/R code/2024-2025/3 EIP Hospitalization Graph_Rates.R")
```

# New report

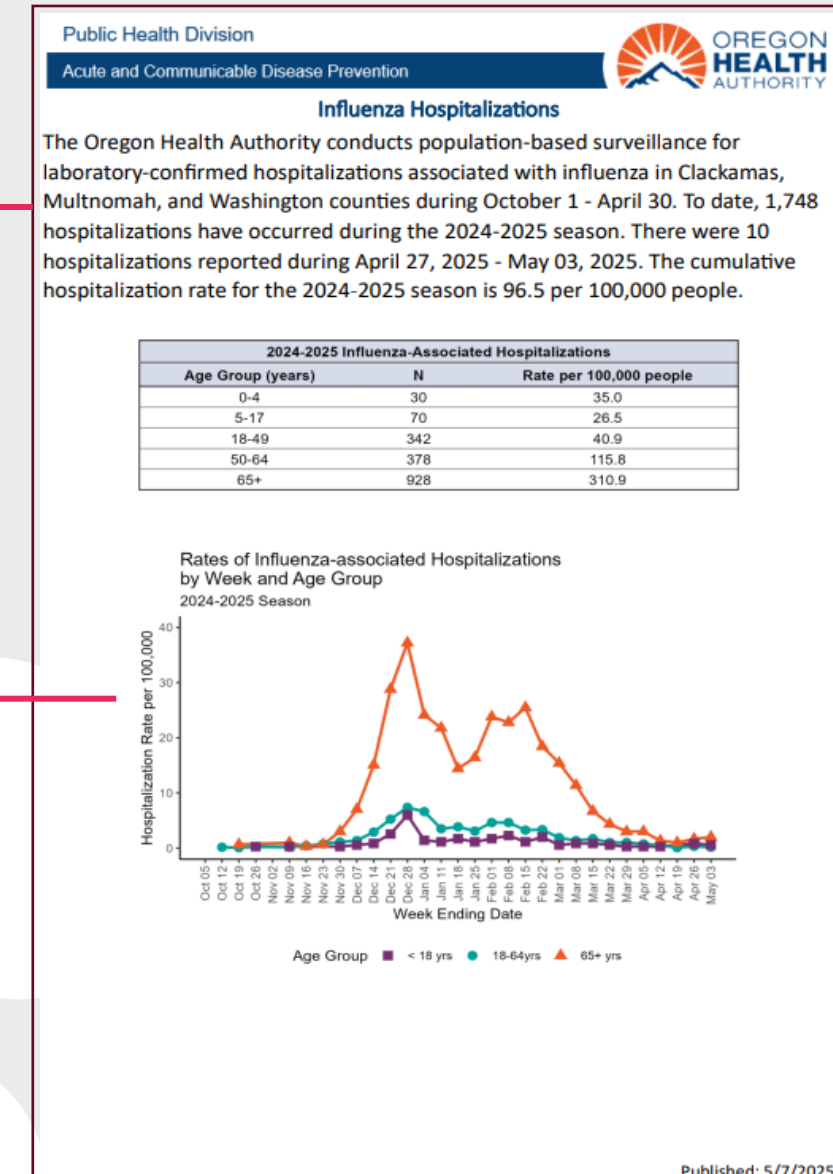


Template with  
agency logo

Fixed text with  
referenced R  
object values

flextable:  
summary of  
laboratory data

geom\_sf:  
Heatmap of  
positive tests

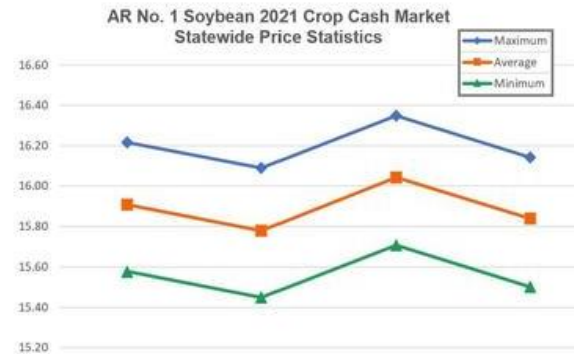


ggplot:  
hospitalization  
rates by week  
and age group

# Accessibility considerations

## Example 1 - Shape Differences

The colored lines also have shape differences (diamond, square, triangle) to identify them.



Harvard digital  
accessibility  
guide

Hemmingway  
App readability  
analysis tool

Leonardo  
color analysis  
tool

| PREVIEW | VISION TYPE   | STATUS | COLOR DIFFERENCE |
|---------|---------------|--------|------------------|
|         | Normal vision | Safe   | Delta E 26.85    |
|         | Deuteranopia  | Unsafe | Delta E 9.53     |
|         | Protanopia    | Unsafe | Delta E 3.48     |
|         | Tritanopia    | Safe   | Delta E 44.26    |
|         | Achromatopsia | Unsafe | Delta E 2.96     |

1 of 13 sentences is  
very hard to read.

1 of 13 sentences is  
hard to read.

Find grammar and spelling  
issues with Editor Plus.  
[Upgrade](#)

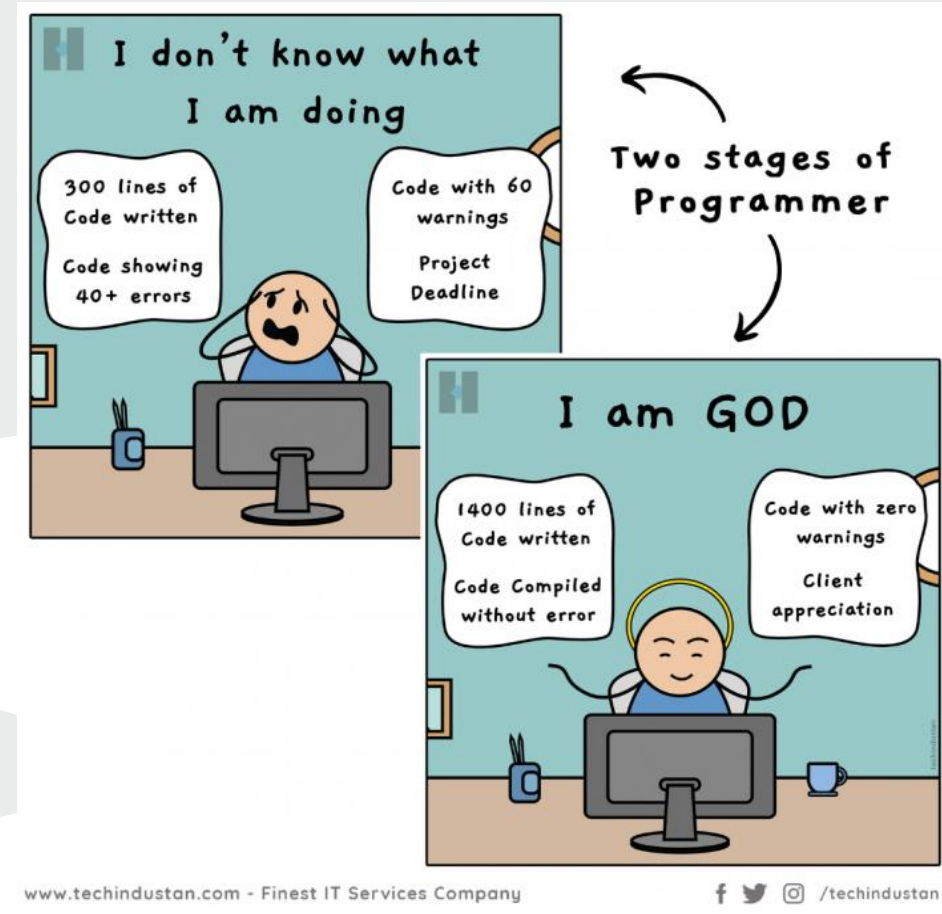
2 weakeners.  
[View details](#)

1 word with a simpler  
alternative.



# Overcome troubleshooting

- **Tapping into the community**- engaging with coworkers for creative solutions and seeking feedback on tables and figures
- **Peer review for robust code**- more people reviewing code helped catch subtle bugs, improve accuracy and reproducibility
- **Building logic checks**- implement automated checks for missing, inconsistent or delayed data
- **"First week wiggles"**-expect some initial hiccups-early runs often reveal unexpected issues, so plan for time reviewing code and fine-tuning



# Return on investments

- **Time saving** - automated workflow decreased our manual reporting from 1 hour per week to 5 minutes.
- **Decrease human error**-consolidating report process eliminates the need to use multiple programs and save multiple Excel spreadsheets.
- **Improve data management**-new reports allow us to backlog hospitalization data and laboratories' weekly tests and test positives.
- **Reuseable framework**-once report has been created, it can be easily used for other projects.
- **Staff cross coverage**-standardized workflow allows for easy coverage when team members are absent.



# Future plans

- Build API connection from REDCap to R for streamlined data, a one-step process for respiratory reports
- Expand automated reporting to additional surveillance reports
  - Yearly hospitalization reports
  - Fax templates
- Create internal Quarto Dashboard to summarize surveillance metrics
  - Display weekly trends for respiratory hospitalizations
  - Support timely, data-driven public health interventions



# Acknowledgments

- Oregon Health Authority R User Group
- Charlotte Wickham, 2024 Quarto Instructor
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- Colin Donald, DEQ

